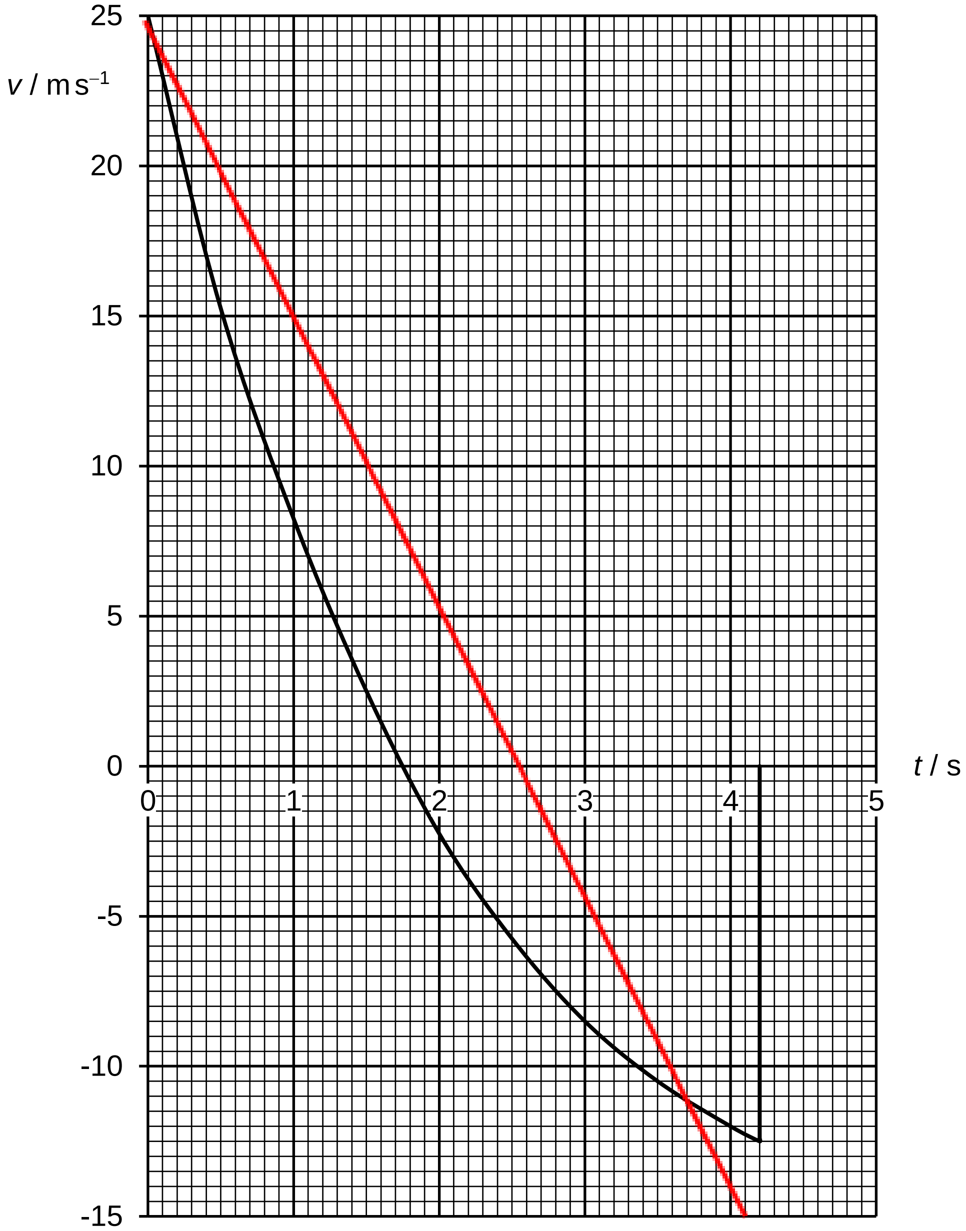
1	(a)	The gradient of the graph represents the acceleration of the ball, which is changing with speed. This suggests that the resultant force acting on the ball (which is proportional to the acceleration by Newton's second law) is changing with speed. When the ball is rising, the resultant force is given by weight and viscous force. Since its weight is constant, a changing resultant force implies that air resistance acting on the ball must vary with speed. OR when the ball is falling, the resultant force = weight – viscous force. A decreasing resultant force will imply that the viscous force is increasing with increase in speed.		**B1** **B1**
		Marker's comment Students should start the explanation by connecting the gradient of v-t graph to acceleration. This will allow the examiner to better follow the discussion. Students tend to be unclear in which part of the motion they were describing. The students did not seem to recognise that while the gradient is decreasing and hence the acceleration is decreasing, the implication is different when the ball is rising compared to the case when the ball is falling. Some answers are rather convoluted and answering directly to the question. Students should state clearly		
	(b)	(i)	time = 1.75 s	**A1**
		Marker's comment Some students did not understand at the highest point, the resultant force is the weight and thus this is the time at which the acceleration of the ball is *g*. There were several students who gave the answers as 4.2 s.		

(ii)



[B1] – Straight line graph starting from (0, 25) with negative gradient.

[B1] – Cuts time axis at t = 2.55 s (so that the gradient of graph is –9.81 m s^{-2})

B2

Marker's comment

Students need to plot the graph to half a small square. It should be shown clearly that the line cuts in between 2.5 and 2.6 s. Students do not gain credit if the graph is not plotted to half a small square.

This part was poorly done as students did not show understanding that the gradient of graph needs to be 9.81 m s^{-2}.

Most students understood that the ball starts with a speed of 25. m s^{-1}

	(c)	For the upward motion, energy lost = loss in KE - gain in GPE energy lost $= \frac{1}{2}\times 0.350\times 25^2 - (0.350\times 9.81\times 19)$ $= 44.14$ J	**C1**
		For the downward motion, energy lost = loss in GPE - gain in KE energy lost $= 0.350\times 9.81\times 19 - \left(\frac{1}{2}\times 0.350\times 12.5^2\right)$ $= 37.89$ J	**C1**
		Therefore, ratio $= \frac{44.14}{37.89}$ ≈ 1.16	**A1**
		Marker's comment Not well done. Students did not make sense of the given data from the graph. Also, some students tried to find the acceleration from the graph which would only be useful if the question assesses on the concepts of dynamics. Students need to connect the graph to the physical motion of the ball to better solve the question.	

2	**(a)**	Moment of a force is the product of the force and the perpendicular distance to a pivot	**B1**